

AI-Assisted Federal Survey Harmonization

Research Fact Sheet

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The Census Bureau alone fields 47 demographic surveys containing ~7,000 questions. AI can map where those surveys overlap and how compatible they are, unlocking new analytical power without collecting a single additional data point.

What’s the problem?

The Census Bureau fields **47 demographic surveys** containing approximately **7,000 questions**, each designed for a specific analytical purpose. No complete map exists showing where these surveys overlap, where their questions are compatible, and where the differences matter. Building that map manually — collecting instruments, systematically comparing question pairs, classifying harmonization barriers, and documenting findings — represents months of effort for a skilled research team. This research accomplished that work in weeks using AI-assisted methods, at a total API cost under \$100.

Why should leadership care?

Response rates are declining across surveys. Every year, it costs more to collect less. Meanwhile, the Census Bureau already *has* enormous data that could answer questions no single survey can — if the connections between surveys were mapped and quality-scored.

The primary value is **data enrichment**: using overlapping questions as bridge variables to link surveys into richer analytical datasets. A secondary value is identifying where **instrument consolidation** could reduce respondent burden. Enrichment gives every agency *more* capability. Nobody loses anything.

How is this different?

Traditional survey harmonization is manual, slow, and limited to 2–3 surveys at a time. This research uses **multiple large language models from different vendors independently rating question pairs**, using multiple models to avoid single-model bias. The approach holds the *entire set of 47 surveys* in view simultaneously, finding connections no individual expert could surface — not because experts lack intelligence, but because no human holds 7,000 questions in working memory at once.

What does the evidence show?

Validation Check	Result	Interpretation
Do the AI models agree with each other?	3 independent models, near-perfect agreement	Task is well-defined; results aren’t driven by any single model’s bias
Are rates consistent across surveys?	CPS→ACS: 42.5%; FoodAPS→ACS: 48.6%	Stable rates suggest a general property, not a survey-specific artifact
Do easy questions score easy?	Race, age, relationship: near-100% consolidability	Framework has face validity
Does the framework explain <i>why</i> things fail?	Construct mismatch dominates failures (~96%)	Correctly identifies barrier types, not just pass/fail
Is the work reproducible?	Full pipeline, data, and code are public	Any researcher can verify results
Is it cost-effective?	Under \$100 total API cost for 1,598 pair analysis	Orders of magnitude cheaper than equivalent manual review

Key findings

~45% of questions across two major survey pairs (CPS and FoodAPS vs. ACS) have at least one viable harmonization path:

- **Tier 1 — Direct recode** (~16%): Close enough to link with simple recoding. High-confidence bridges.
- **Tier 2 — Statistical adjustment** (~29%): Linkable with documented transformations. Usable bridges with known limitations.
- **Tier 3 — Not linkable** (~55%): Different constructs. The framework correctly identifies these and explains why.

What's at stake?

Without this work, these 47 surveys continue as isolated silos. Declining response rates erode quality with no compensating strategy. Cross-survey analytical opportunities, connections that could inform policy without additional collection, remain invisible. The methodology demonstrated here could extend to surveys across the broader federal statistical system.

What's next?

Report 04 extends the analysis from pairwise comparisons to the full multi-survey network, discovering multi-hop enrichment paths (Survey A → B → C) across all 47 Census demographic surveys — connections that exceed any individual researcher's working memory.

DRAFT — These results are preliminary and have not been independently validated or peer reviewed. Full methodology, data, and code: github.com/brockwebb/federal-survey-concept-mapper. The views expressed are the author's own and do not necessarily represent the views of the U.S. Census Bureau or the U.S. Department of Commerce.